



Genes and Genomes

Dr Saroor Patel

Wellcome Trust Sanger
Institute



Overview

- Genes and Genomes
 - Genome assemblies
 - Gene annotation
 - Gene regulation
- Genome Browsers
 - Ensembl
 - UCSC
- CRISPR Tools



Genes and Genomes

- Genome assemblies
 - Genome Reference Consortium releases include major and minor updates (patches)
 - Human, Mouse and Zebrafish reference genomes 'finished' to the highest standard
 - Telomere-to-Telomere (T2T) provides first complete assembly
 - Other genomes based on 'unfinished' de-novo assemblies
- Gene builds
 - Based on specific genome assemblies
 - Regular releases for data updates

Genome Reference Consortium (GRC)

- Goals

- Responsible for maintaining human and mouse reference genomes
- To correct regions in the genome that are currently misrepresented
- To close as many genomic gaps as possible
- To produce alternative assemblies of structurally variant loci where necessary
- Scientific community can report loci in need of review



<https://www.ncbi.nlm.nih.gov/grc>

GRC – human genome assemblies

	GRCh38.p14	GRCh38.p5	GRCh38	GRCh37	NCBI36
Release Date	February 03, 2022	September 25, 2015	December 24, 2013	February 27, 2009	March, 2006
General Info					
Assembly Type	Haploid with alt loci	haploid with alt loci	haploid with alt loci	haploid with alt loci	haploid with alt loci
Release Type	patch	patch	major	major	major
Number of Assembly Units	38	38	37	10	6
Total Bases in Assembly	3,298,912,062	3,230,538,373	3,209,286,105	3,137,144,693	3,104,054,490
Total Non-N Bases in Assembly	3,137,300,923	3,069,170,222	3,049,316,098	2,897,299,566	2,881,649,121
Primary Assembly NSO	67,794,873	67,794,873	67,794,873	46,395,641	38,509,590
Region Information					
Total number of defined regions	434	236	207	7	8
Number of Regions with Alternate Loci	178	178	178	3	0
Number of Regions with Fix Patches	164	37	0	0	0
Number of Regions with Novel Patches	90	21	0	0	0
Number of Regions as PAR	4	4	4	4	8
Alternate Loci/PATCH Information					
Total Number of Alternate Loci scaffolds	261	261	261	9	11
Number of Alternate Loci scaffolds aligned to the Primary Assembly	261	261	261	9	0
Number of FIX Patch scaffolds	164	38	0	0	0
Number of FIX Patch scaffolds aligned to the Primary Assembly	161	37	0	0	0
Number of NOVEL Patch scaffolds	90	24	0	0	0
Number of NOVEL Patch scaffolds aligned to the Primary Assembly	90	24	0	0	0

GRC – mouse genome assemblies

	GRCm39	GRCm38.p6	GRCm38.p2	GRCm38	MGSCv37
Release date	June 24, 2020	September 15, 2017	March 15, 2013	January 9, 2012	September 23, 2010
General Info					
Assembly Type	haploid	haploid with alt loci	haploid with alt loci	haploid with alt loci	haploid with alt loci
Release Type	major	patch	patch	major	major
Number of Assembly Units	2	17	17	16	15
Total Bases in Assembly	2,728,222,451	2,818,974,548	2,798,785,524	2,793,712,140	2,745,142,291
Total Non-N Bases in Assembly	2,654,621,837	2,739,538,976	2,719,482,043	2,714,422,680	2,648,522,751
Primary Assembly N50	106,145,001	54,517,951	54,517,951	54,517,951	39,293,786
Region Information					
Total number of defined regions	3	129	83	72	60
Number of Regions with Alternate Loci	0	70	70	70	60
Number of Regions with Fix Patches	0	65	14	0	0
Number of Regions with Novel Patches	0	9	0	0	0
Number of Regions as PAR	2	2	2	2	0
Alternate Loci/PATCH Information					
Total Number of Alternate Loci scaffolds	0	99	99	99	94
Number of Alternate Loci scaffolds aligned to the Primary Assembly	0	92	92	92	76
Number of FIX Patch scaffolds	0	65	14	0	0
Number of FIX Patch scaffolds aligned to the Primary Assembly	0	64	14	0	0
Number of NOVEL Patch scaffolds	0	9	0	0	0
Number of NOVEL Patch scaffolds aligned to the Primary Assembly	0	9	0	0	0

How do genome assemblies effect CRISPRs?

- Any differences between reference genome assemblies may:
 - Add or remove CRISPR sequences in databases
 - Effect off-target analysis

How do genome assemblies effect CRISPRs?

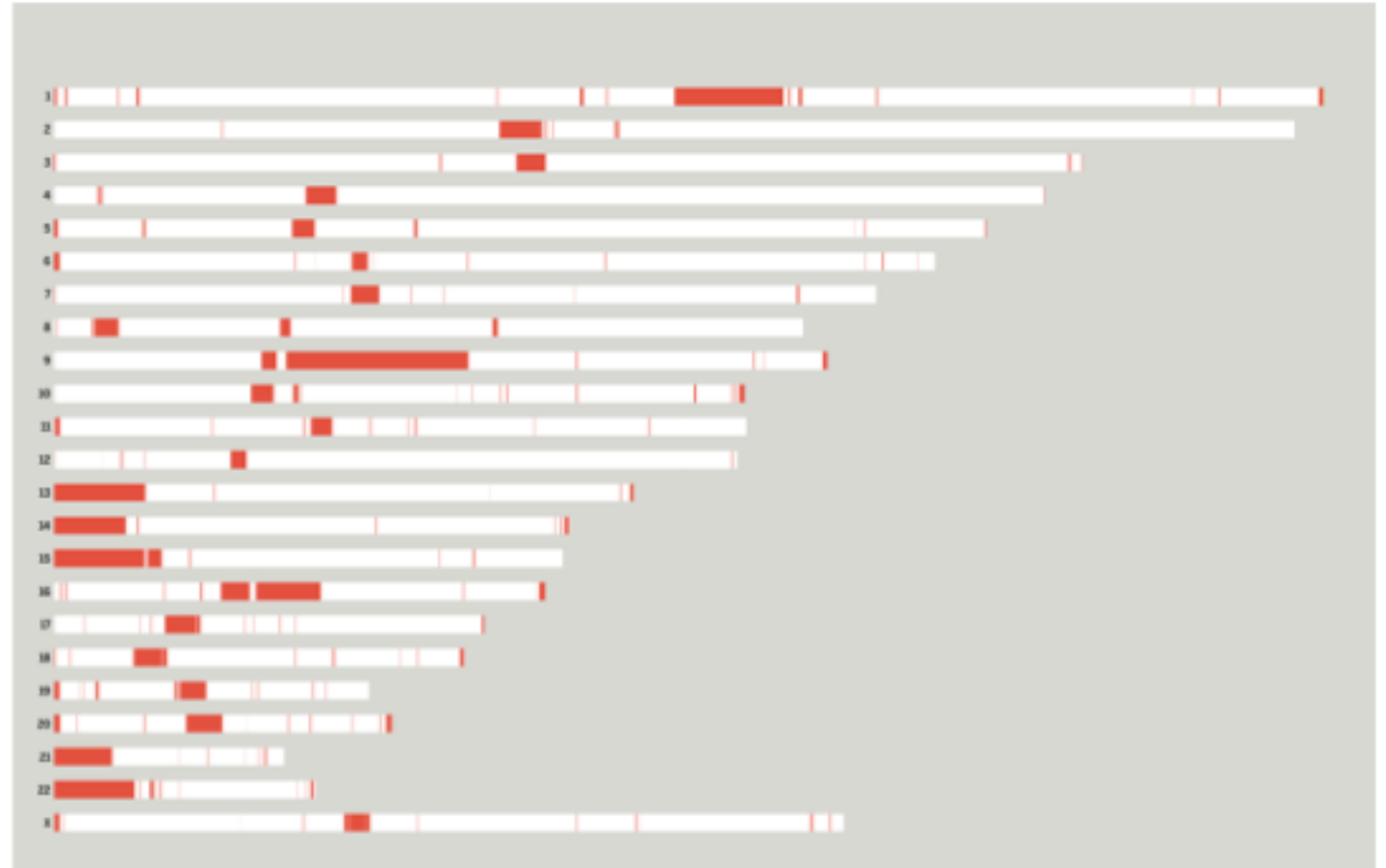
- Any differences between reference genome assemblies may:
 - Add or remove CRISPR sequences in databases
 - Effect off-target analysis
- Example: TGTCTCAGGTCCCGCATGAT TGG

	0	1	2	3	4	Total
MGSCv37 (mm9)	1	0	1	9	104	114
GRCm38 (mm10)	12	141	670	1166	3020	5009

Telomere-to-Telomere (T2T) Consortium



<https://www.science.org/doi/10.1126/science.abj6987>



Non-reference Genomes

- Personal genomes
 - Patient-derived iPS cells, therapeutics
- Cell Lines
 - CNVs, polyploidy
- Whole Genome Duplications
 - Eg. Zebrafish
- Non-reference Genomes
 - Haplotypes and/or strain differences

Non-reference Genomes

[illegible]

Gene Annotation

- RefSeq
 - RefSeq transcript and protein records are generated by several processes including:
 - Computation (mRNA XM_1234; Protein XP_1234; RNA XR_1234)
 - Manual curation (mRNA NM_1234; Protein NP_1234; RNA NR_1234)
 - Annotation includes 392 eukaryotic genomes. **Transcript-centric.**
- Ensembl
 - Computational alignment-based annotation, including cDNAs, proteins and RNA-seq reads for over 70 different vertebrate species. **Genome-centric.**

Gene Annotation

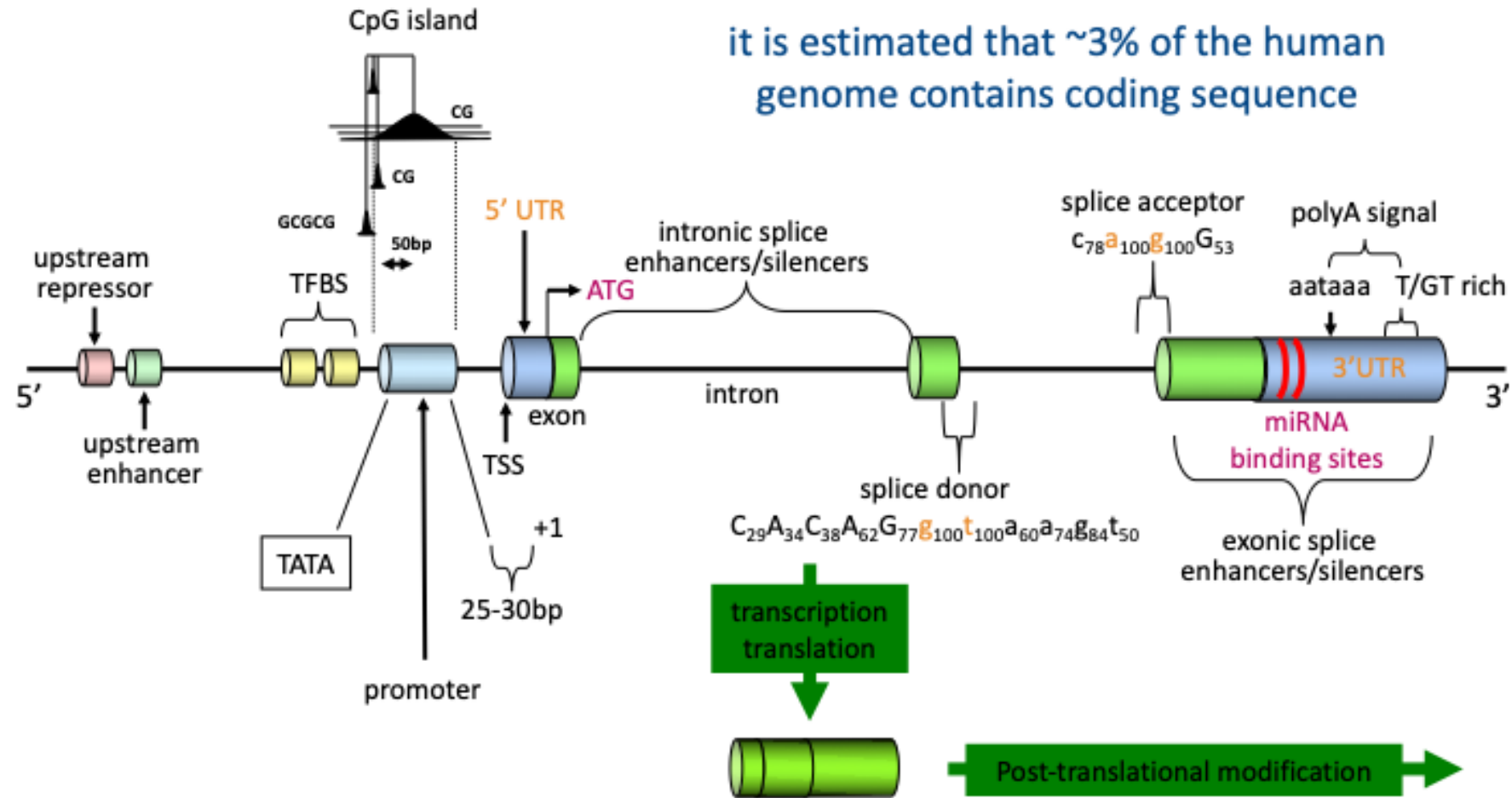
- GENCODE

- High quality reference gene annotation and experimental validation for human and mouse genomes, based on manual annotation that includes Ensembl annotation.

- MANE

- **M**atched **A**nnotation between **N**CBi and **E**nsembl
- Exact match between a Refseq transcript and Ensembl/GENCODE annotation such that the two identifiers can be used synonymously.
- Matches GRCh38 reference genome assembly perfectly and is chosen based on biologically relevant criteria, such as transcript expression levels and conservation of coding regions.

Gene structure



UnTranslated Regions (UTR)

Different 5' UTR isoforms associated with:

- Promoters (eg. tissue specificity)
- Transcription Start Sites (TSS)
- Translation Start Sites (ATG - Methionine)

Differences in the 3' UTR associated with:

- Alternative polyAdenylation (ApA)
- mRNA stability, localisation, expression and regulation

Identifying Transcription Start Sites (TSS)

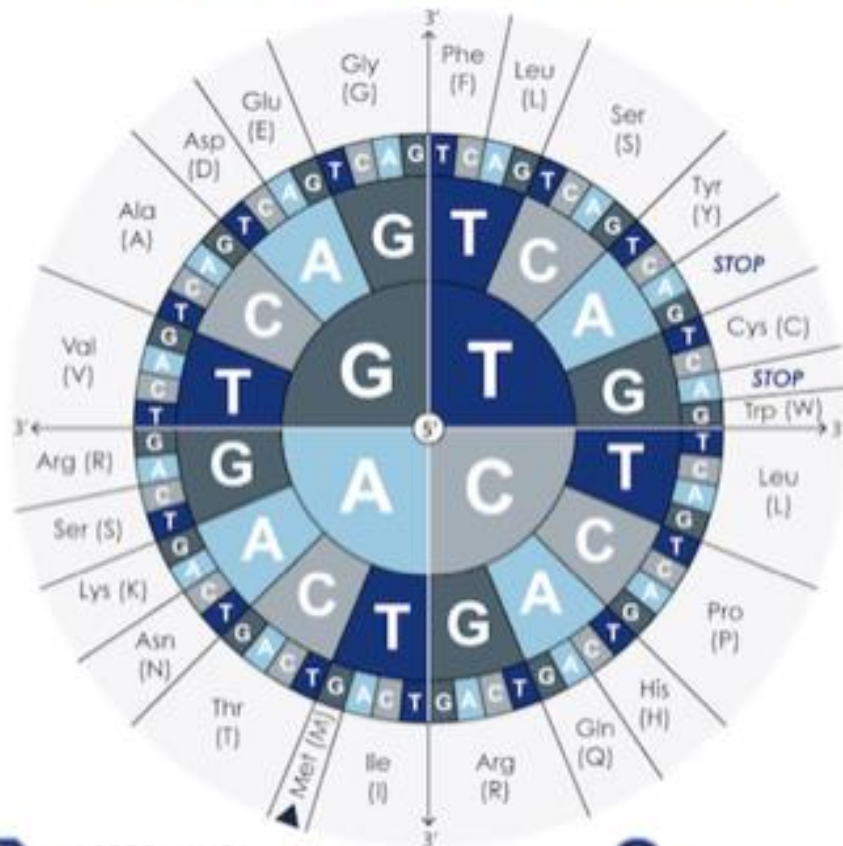
- GENCODE TSS positions coincide with the first nucleotide position of transcripts
 - often based on extent of evidence, rather than experimental validation
- Large scale experimental data provided by the ENCODE and Fantom5 consortiums
 - ENCODE contains CAGE and RAMPAGE data
 - Fantom5 provides CAGE-seq peaks identifying TSS in different tissues and cell types for human and mouse
- Ongoing efforts to improve TSS annotation, with associated experimental data.

Coding Sequence

- Coding sequence (CDS) defines translated regions.
- Different transcripts can have a different CDS for the same gene.
- Every three bases of a CDS defines a codon for a specific tRNA.
- Provides 3-frames of translation for forward or reverse strand.
- The same exon may encode different CDS, depending on alternative splicing.

Coding Sequence and Codons

cDNA Genetic Code



MASTERMIND®
THE GENOMIC SEARCH ENGINE

Sign up for Mastermind
<http://bit.ly/mastermind-codon>



GENOMENON®

cDNA Codon Table

		Second Position				Third Position
First Position		T	C	A	G	
	T	TTT } Phe TTC } TTA } Leu TTG }	TCT } Ser TCC } TCA } TCG }	TAT } Tyr TAC } TAA } STOP TAG } STOP	TGT } Cys TGC } TGA } STOP TGG } Trp	T C A G
	C	CTT } Leu CTC } CTA } CTG }	CCT } Pro CCC } CCA } CCG }	CAT } His CAC } CAA } Gln CAG }	CGT } Arg CGC } CGA } CGG }	T C A G
	A	ATT } Ile ATC } ATA } Met ATG }	ACT } Thr ACC } ACA } ACG }	AAT } Asn AAC } AAA } Lys AAG }	AGT } Ser AGC } AGA } Arg AGG }	T C A G
	G	GTT } Val GTC } GTA } GTG }	GCT } Ala GCC } GCA } GCG }	GAT } Asp GAC } GAA } Glu GAG }	GGT } Gly GGC } GGA } GGG }	T C A G



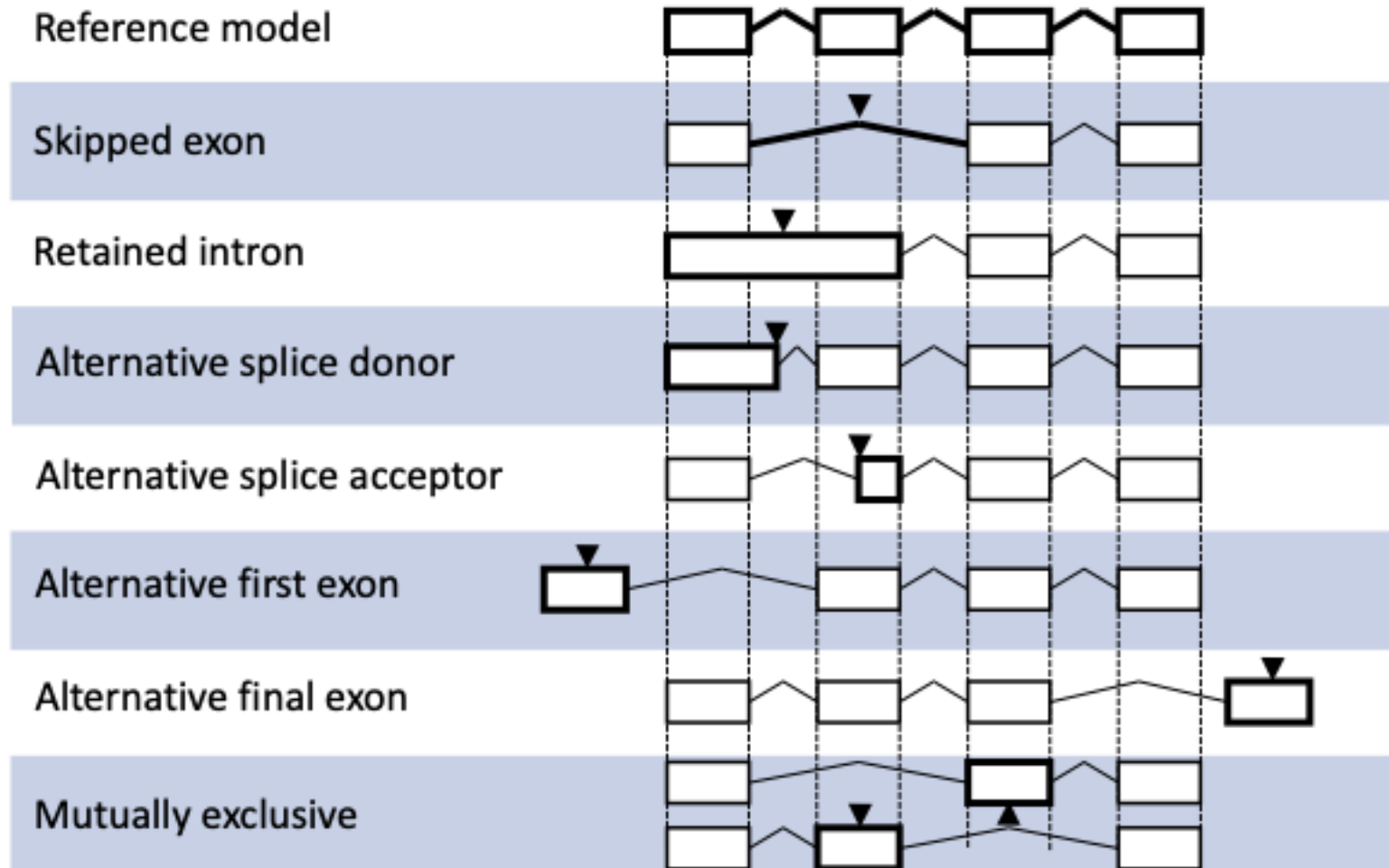
MASTERMIND®
THE GENOMIC SEARCH ENGINE

www.genomenon.com
734-794-3075
help@genomenon.com



GENOMENON®

Alternative splicing



Nonsense mediated decay (NMD)

Cellular mechanism that destroys potentially harmful splice variants.

Coding transcript model



NMD transcript model



NMD transcript model



NMD transcript model

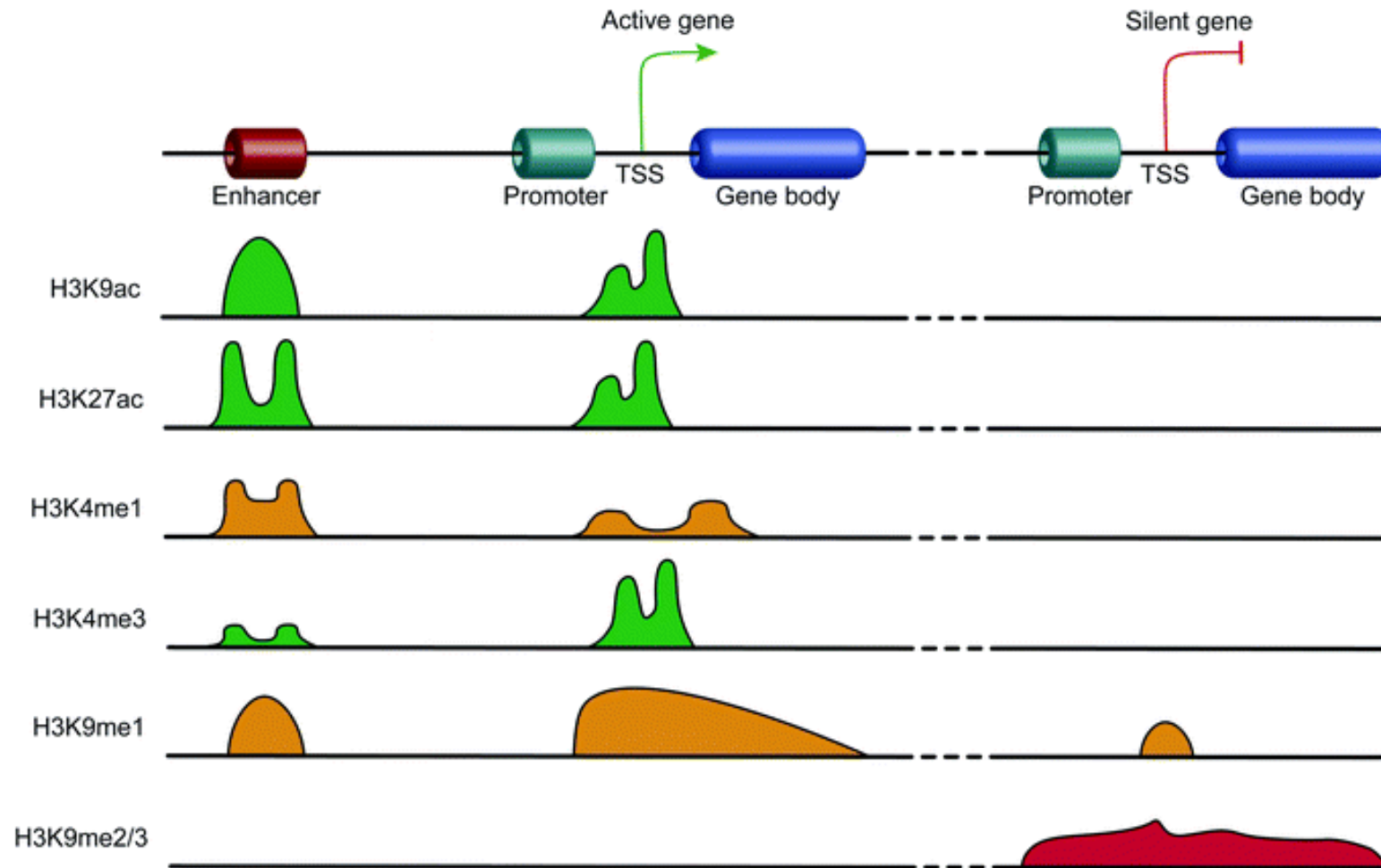


NMD transcript model



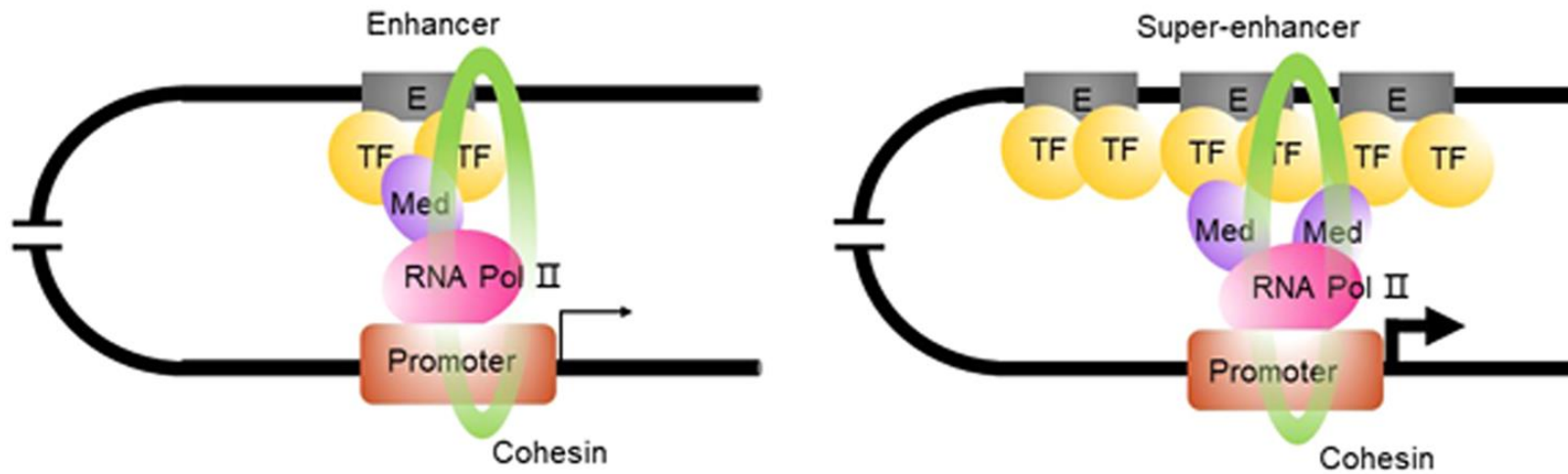
* Requires premature termination codon >50bp upstream of the final splice site.

Gene regulation: enhancers



From Musolino et al. Nanoscale Adv 2022

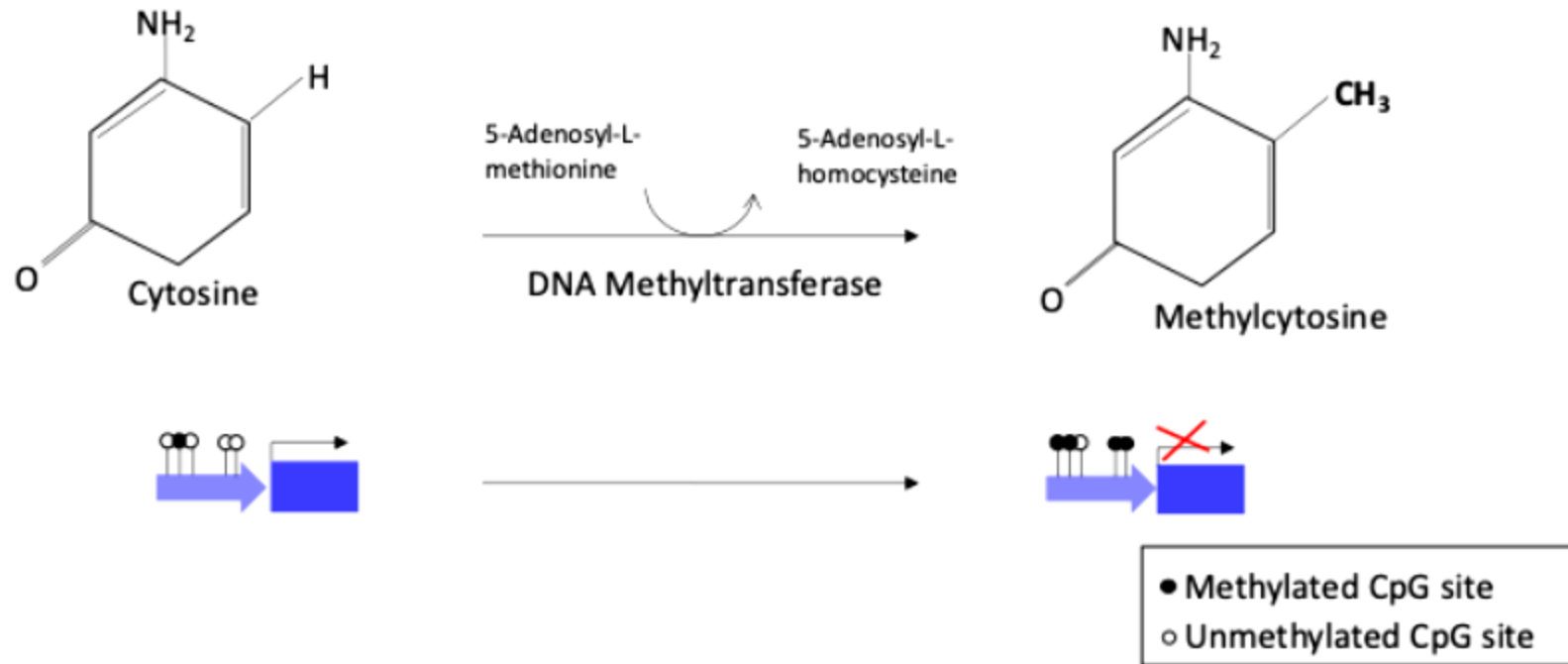
Gene regulation: enhancers



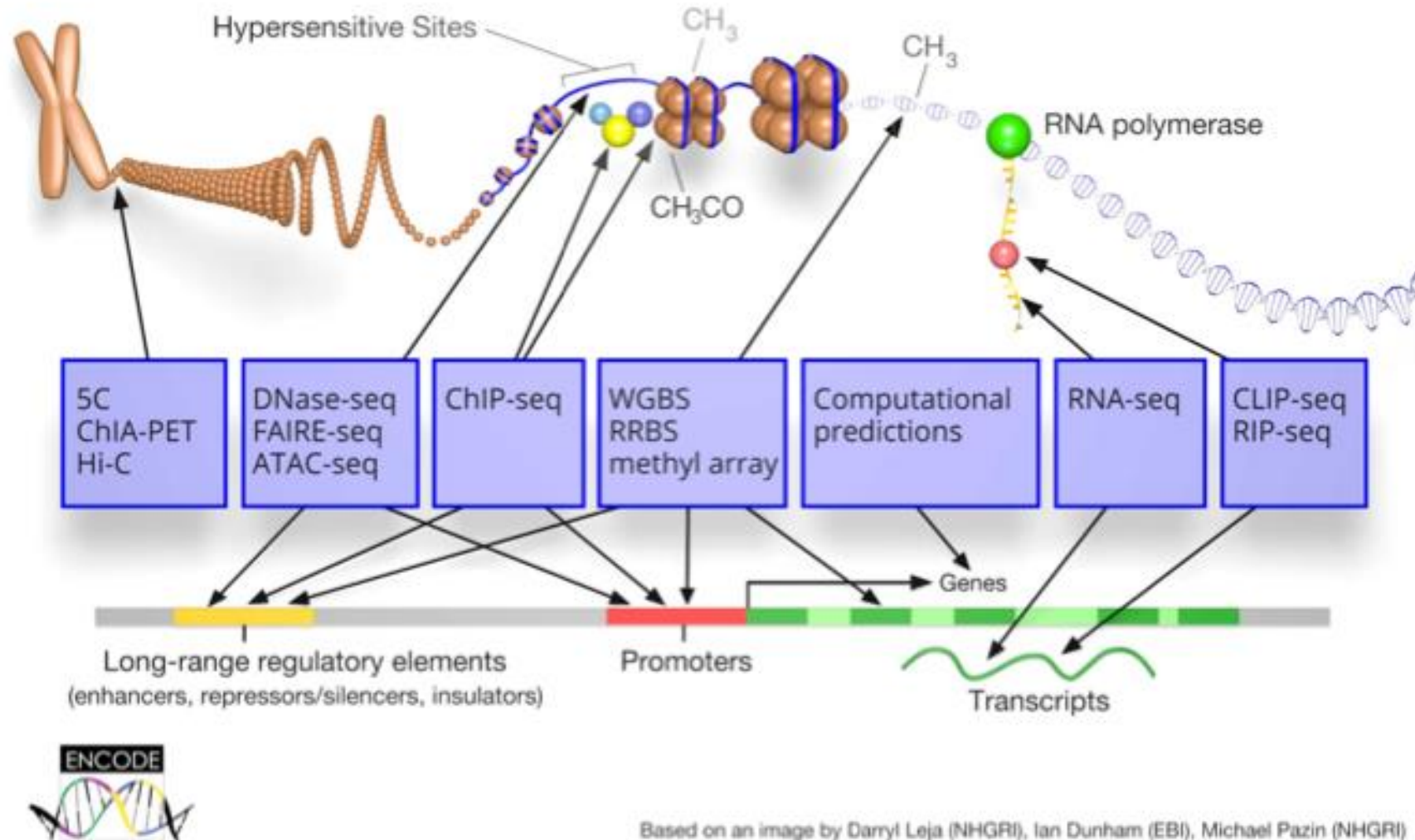
From Jia et al. Exp & Mol Medicine 2020

Gene regulation: CpG islands and DNA methylation

Often found near promoter regions, when methylated typically inactivates gene expression



ENCODE: Encyclopedia of DNA Elements



Based on an image by Darryl Leja (NHGRI), Ian Dunham (EBI), Michael Pazin (NHGRI)

ENCODE: Encyclopedia of DNA Elements

ENCODEDataEncyclopediaMaterials & MethodsHelp

Search...

Ground Level Annotations

Gene expression (RNA-seq)

The expression levels of genes and transcripts annotated by GENCODE in over 200 human and 90 mouse experiments.

[Long RNA-seq Data](#) | [Query](#) | [Download](#) | [Method](#)



BRCA1 Gene Expression

Transcription factor binding sites (TFBS)

Peaks (enriched genomic regions) of transcription factors.

[Raw Data](#) | [Peaks](#)

Visualize sequence motifs and other annotations.

Open chromatin (DNase-seq)

DNase I hypersensitive sites (also known as DNase-seq peaks) computed from ~300 human and mouse experiments.

[Raw Data](#) | [Peaks](#)

Topologically associating domains (TADs) and compartments (Hi-C)

TADs and A and B compartments computed from 12 human cell lines.

[Raw Data](#) | [Visualize](#)

Promoter-enhancer links (ChIA-PET)

Links between promoters and distal regulatory elements such as enhancers computed from 8 ChIA-PET experiments.

[Raw Data](#) | [Links](#)

RNA binding protein occupancy (eCLIP-seq)

Peaks computed from eCLIP-seq data in human cell lines K562 and HepG2 for a large number of RNA Binding Proteins (RBPs).

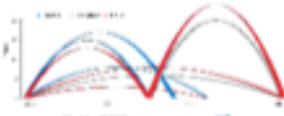
[Raw Data](#) | [Peaks](#)



CTCF DHS Profile



K562 Interaction Matrix



ChIA-PET Interactions



RBP eCLIP

RBP read density

encodeproject.org

ENCODE: Encyclopedia of DNA Elements

- ENCODE data can be displayed in both the Ensembl and UCSC browsers (default).
- Data is mapped to the reference genome assemblies.
- Not all data is mapped to the newest reference assemblies ie. GRCh38 (hg38), GRCm38 (mm10)
 - availability of browser tracks therefore differs between builds
- Released data can be downloaded.

Genome Browsers

- Ensembl
 - provides tools for viewing and analyzing genomic data
 - includes annotation of regulatory, transcriptional and variant data
 - <http://www.ensembl.org>
- UCSC
 - provides tools for viewing and analyzing genomic data
 - <http://genome.ucsc.edu>

Genome browser: Ensembl

Go to <http://www.ensembl.org> and search for the Apoe gene.

 [BLAST/BLAT](#) | [VEP](#) | [Tools](#) | [BioMart](#) | [Downloads](#) | [Help & Docs](#) | [Blog](#)

[Login/Register](#)

[Tools](#)
[All tools](#)

[BioMart >](#)
Export custom datasets from Ensembl with this data-mining tool

[BLAST/BLAT >](#)
Search our genomes for your DNA or protein sequence

[Variant Effect Predictor >](#)
Analyse your own variants and predict the functional consequences of known and unknown variants

Search

for

Go

e.g. BRCA2 or ref S-62797383-63627989 or rs699 or coronary heart disease

All genomes

 **Pig breeds**
Pig reference genome and 12 additional breeds
[View full list of all species](#)

Favourite genomes 

 **Human**
GRCh38.p14
[See also GRCp11](#)

 **Mouse**
GRCm38

 **Zebrafish**
GRCz11

Ensembl is a genome browser for vertebrate genomes that supports research in comparative genomics, evolution, sequence variation and transcriptional regulation. Ensembl annotate genes, computes multiple alignments, predicts regulatory function and collects disease data. Ensembl tools include BLAST/BLAT, BioMart and the Variant Effect Predictor (VEP) for all supported species.

Ensembl Release 112 (May 2024)

- Many new fish genomes have been added to Ensembl
- Population frequency data is available for chicken, dog, goat and sheep through VEP
- Update to our current regulation annotation. The promoters now align with the 5' ends of known transcripts
- VEP will be updated to use the dbNSFP commercial data release

[More release news >](#) on our blog

Ensembl Rapid Release
New assemblies with gene and protein annotation every two weeks
Note: species that already exist on this site will continue to be updated with the full range of annotations
[Go](#)
The Ensembl Rapid Release website provides annotation for recently produced, publicly available vertebrate and non-vertebrate genomes from biodiversity initiatives such as Darwin Tree of Life, the Vertebrate Genomes Project and the Earth BioGenome Project
[Rapid Release news >](#) on our blog

[Compare genes across species](#)


[Find SNPs and other variants for my gene](#)


[Gene expression in different tissues](#)


[Retrieve gene sequence](#)


[Find a Data Channel](#)


[Use my own data in Ensembl](#)


Genome browser: Ensembl

The results table lists "ApoE" found in all species, including entries for gene and transcripts.

The screenshot shows the Ensembl genome browser interface. At the top, the Ensembl logo is followed by navigation links: BLAST/BLAT, VEP, Tools, BioMart, Downloads, Help & Docs, and Blog. Below this is a 'New Search' button. The main content area is divided into two columns. The left column contains two filters: 'Restrict category to:' and 'Restrict species to:'. The 'Restrict category to:' filter shows a list of categories with their respective counts: Gene (548), Transcript (316), Variant (1485), Phenotype (60), GeneTree (88), GenomicAlignment (66437), ProbeFeature (7), and Protein Domain (945). The 'Restrict species to:' filter shows a list of species with their respective counts: Human (2365), Mouse (9733), Zebrafish (579), Abingdon island giant tortoise (8), African green monkey (7), and African ostrich (1). A blue arrow points from the 'Human' entry in the 'Restrict species to:' filter to the 'APOE-201 (Human Transcript)' entry in the results table. The right column shows the search results for 'apoe'. It starts with a search bar containing 'apoe' and a magnifying glass icon. Below the search bar, it says '69886 results match apoe'. The first result is 'APOE (Human Gene)' with the Ensembl ID 'ENSG00000130203' and a description 'Apolipoprotein E [Source:HGNC Symbol;Acc:HGNC:613]'. Below this is a description of the gene: 'APOLIPOPROTEIN E: APOE [107741] (MIM gene record; description: APOLIPOPROTEIN E: APOE) is an external reference matched to Gene ENSG00000130203'. Below the description are links for 'Variant table', 'Phenotypes', 'Location', 'External Refs', 'Regulation', 'Orthologues', and 'Gene tree'. The second result is 'APOE carrier status (Human Phenotype)' with the Ensembl ID 'ENST00000252486' and a description 'Apolipoprotein E [Source:HGNC Symbol;Acc:HGNC:613]'. Below this are links for 'Location', 'External Refs', 'cDNA seq', 'Exons', 'Variant table', 'Protein seq', 'Population', and 'Protein summary'. The third result is 'APOE-203 (Human Transcript)' with the Ensembl ID 'ENST00000434152' and a description 'Apolipoprotein E [Source:HGNC Symbol;Acc:HGNC:613]'. Below this are links for 'Location', 'External Refs', 'cDNA seq', 'Exons', 'Variant table', 'Protein seq', 'Population', and 'Protein summary'. The fourth result is 'APOE-202 (Human Transcript)' with the Ensembl ID 'ENST00000425718' and a description 'Apolipoprotein E [Source:HGNC Symbol;Acc:HGNC:613]'. Below this are links for 'Location', 'External Refs', 'cDNA seq', 'Exons', 'Variant table', 'Protein seq', 'Population', and 'Protein summary'.

Restrict category to:

Gene	548
Transcript	316
Variant	1485
Phenotype	60
GeneTree	88
GenomicAlignment	66437
ProbeFeature	7
Protein Domain	945

Restrict species to:

Human	2365
Mouse	9733
Zebrafish	579
Abingdon island giant tortoise	8
African green monkey	7
African ostrich	1
... 268 more species ...	

apoe

69886 results match apoe

[APOE \(Human Gene\)](#)
ENSG00000130203 19-44905791-44909393.1
Apolipoprotein E [Source:HGNC Symbol;Acc:HGNC:613]
APOLIPOPROTEIN E: **APOE** [107741] (MIM gene record; description: APOLIPOPROTEIN E: **APOE**) is an external reference matched to Gene ENSG00000130203
[Variant table](#) • [Phenotypes](#) • [Location](#) • [External Refs](#) • [Regulation](#) • [Orthologues](#) • [Gene tree](#)

[APOE carrier status \(Human Phenotype\)](#)
Human Phenotype
APOE carrier status

[APOE-201 \(Human Transcript\)](#)
ENST00000252486 19-44905796-44909393.1
Apolipoprotein E [Source:HGNC Symbol;Acc:HGNC:613]
[Location](#) • [External Refs](#) • [cDNA seq](#) • [Exons](#) • [Variant table](#) • [Protein seq](#) • [Population](#) • [Protein summary](#)

[APOE-203 \(Human Transcript\)](#)
ENST00000434152 19-44905812-44909025.1
Apolipoprotein E [Source:HGNC Symbol;Acc:HGNC:613]
[Location](#) • [External Refs](#) • [cDNA seq](#) • [Exons](#) • [Variant table](#) • [Protein seq](#) • [Population](#) • [Protein summary](#)

[APOE-202 \(Human Transcript\)](#)
ENST00000425718 19-44906360-44908954.1
Apolipoprotein E [Source:HGNC Symbol;Acc:HGNC:613]
[Location](#) • [External Refs](#) • [cDNA seq](#) • [Exons](#) • [Variant table](#) • [Protein seq](#) • [Population](#) • [Protein summary](#)

Genome browser: Ensembl

Click on the Apoe gene entry to open the 'Gene Tab'

Gene: Apoe

Gene: Apoe ENSMUSG00000002985

Description	apolipoprotein E [Source: MGI Symbol; Acc: MGI:88057]
Gene Synonyms	Apoe
Location	Chromosome 7: 19,430,034-19,433,113 reverse strand. GRCm39: CM001000.3
About this gene	This gene has 11 transcripts (splice variants), 219 orthologues , 3 paralogues and is associated with 190 phenotypes .
Transcripts	Show transcript table

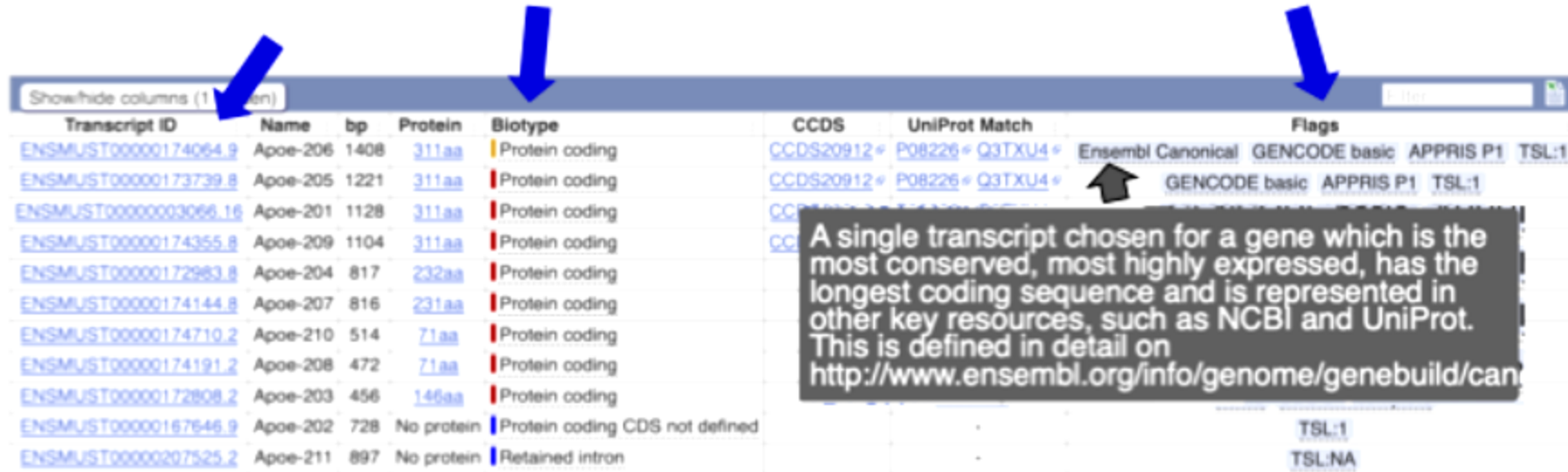
Summary ⓘ

Name	Apoe ⓘ (MGI Symbol)
UniProtKB	This gene has proteins that correspond to the following UniProtKB identifiers: P08226 ⓘ
CCDS	This gene is similar to a CCDS gene on Mouse GRCm39: CCDS20912.1 ⓘ
Ensembl version	ENSMUSG00000002985.17
Gene type	Protein coding
Annotation method	Annotation for this gene includes both automatic annotation from Ensembl and Havana manual curation, see article .

Click on "Show transcript table"

Genome browser: Ensembl

This allows you to view details such as identifiers, biotypes and transcript “Flags”.



Transcript ID	Name	bp	Protein	Biotype	CCDS	UniProt Match	Flags
ENSMUST00000174064.9	Apoe-206	1408	311aa	Protein coding	CCDS20912	P08226 Q3TXU4	Ensembl Canonical GENCODE basic APPRIS P1 TSL:1
ENSMUST00000173739.8	Apoe-205	1221	311aa	Protein coding	CCDS20912	P08226 Q3TXU4	GENCODE basic APPRIS P1 TSL:1
ENSMUST00000003066.16	Apoe-201	1128	311aa	Protein coding	CCDS20912	P08226 Q3TXU4	
ENSMUST00000174355.8	Apoe-209	1104	311aa	Protein coding	CCDS20912	P08226 Q3TXU4	
ENSMUST00000172983.8	Apoe-204	817	232aa	Protein coding			
ENSMUST00000174144.8	Apoe-207	816	231aa	Protein coding			
ENSMUST00000174710.2	Apoe-210	514	71aa	Protein coding			
ENSMUST00000174191.2	Apoe-208	472	71aa	Protein coding			
ENSMUST00000172808.2	Apoe-203	456	146aa	Protein coding			
ENSMUST00000167646.9	Apoe-202	728	No protein	Protein coding CDS not defined			TSL:1
ENSMUST00000207525.2	Apoe-211	897	No protein	Retained intron			TSL:NA

A single transcript chosen for a gene which is the most conserved, most highly expressed, has the longest coding sequence and is represented in other key resources, such as NCBI and UniProt. This is defined in detail on <http://www.ensembl.org/info/genome/genebuild/can>

Position the cursor over a flag to view more information.

Close the transcript table and view the graphical display of all transcripts within the genomic context.

Genome browser: Ensembl

From this view, use the side menu to view data such as supporting evidence, exon information and protein domain.

Transcript-based display

Summary

Transcript: ENSMUST00000174064.9 Apoe-206

Description: apolipoprotein E [Source MGI Symbol Acc: MGI:89357.4]

Gene Synonyms: Apoe

Location: Chromosome 7: 19,436,024-19,436,112 reverse strand

About this transcript: This transcript has 6,888bp, is annotated with 1.6 domains and features, is associated with 655 variant alleles and maps to 258 oligo probes

Gene: This transcript is a product of gene ENSMUSG0000002085.17 [Show transcript table](#)

Summary

[Export image](#)

Apoe-206 ENSMUST00000174064.9 protein-coding

Reverse strand

Statistics: Exons: 4, Coding exons: 3, Transcript length: 1,408 bps, Translation length: 311 residues

Uniprot: This transcript corresponds to the following Uniprot identifiers: [P56226-6](#)

CCDS: This transcript is a member of the Mouse CCDS set: [CCDS20812.6](#)

Transcript Support Level (TSL): [TSL:1](#)

Version: ENSMUST00000174064.9

Type: Protein-coding

Annotation Method: Transcript where the Ensembl genebuild transcript and the Havana manual annotation have the same sequence, for every base pair. See [s0006](#)

GENCODE basic gene: This transcript is a member of the [Genecode basic](#) gene set.

Ensembl release 112 May 2024 © [ENSL, CH](#)

[Permanest link](#) [View in archive](#)

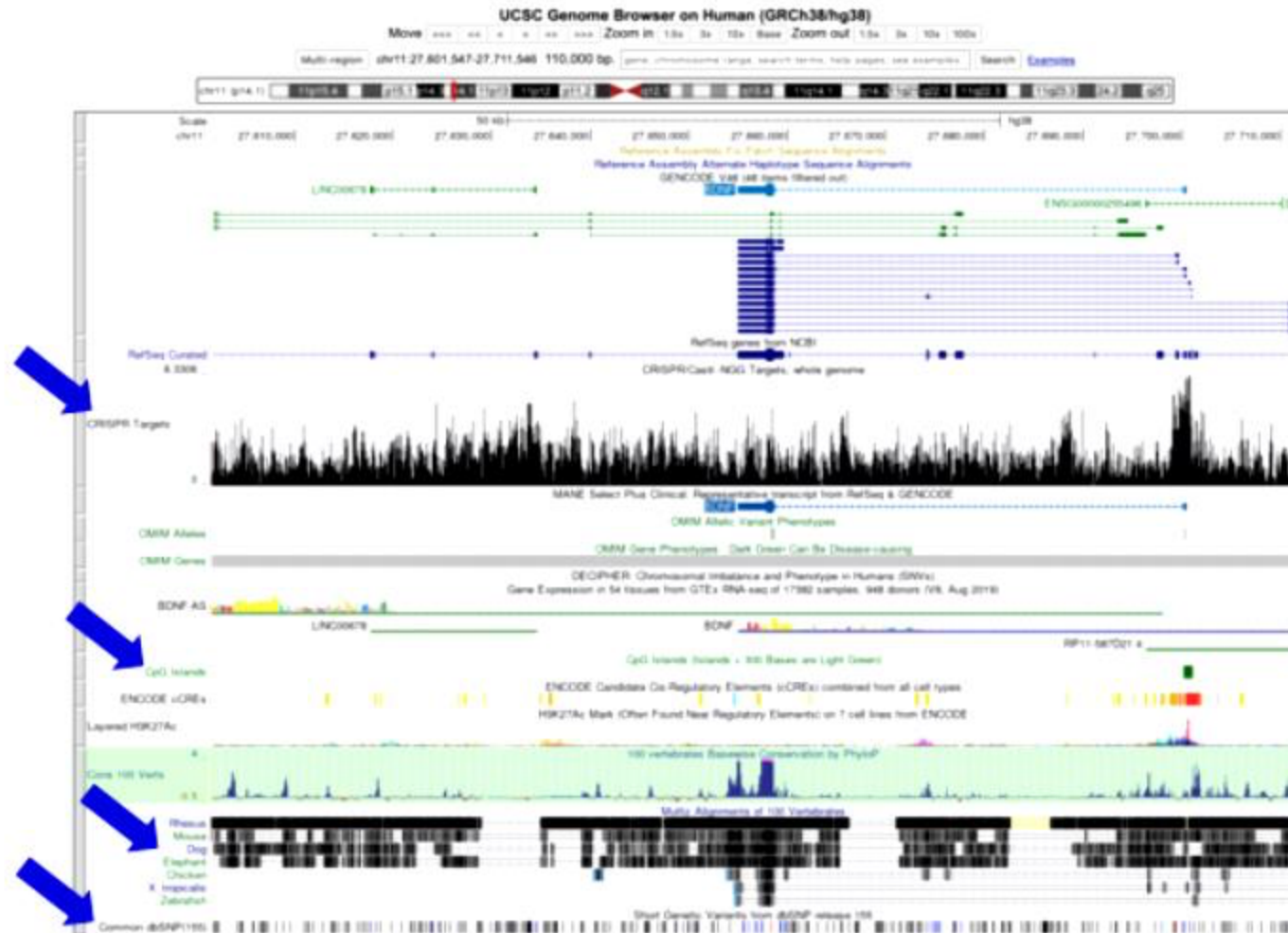
Genome browser: Ensembl

Variation data in Ensembl is a colour coded annotation that denotes the consequence of the change, e.g. frameshift.



Genome browser: UCSC

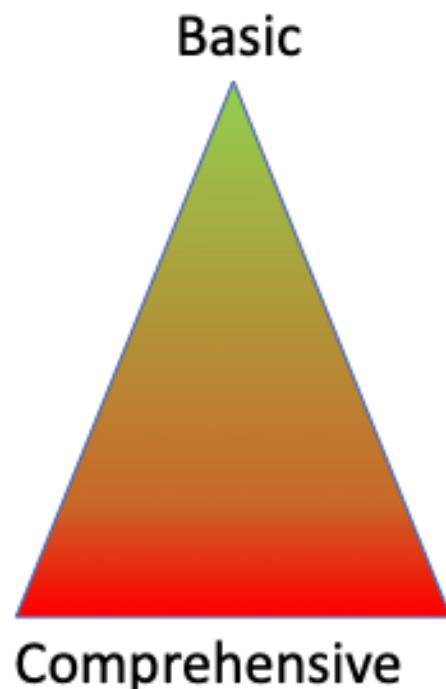
UCSC genome browser: <http://genome.ucsc.edu>



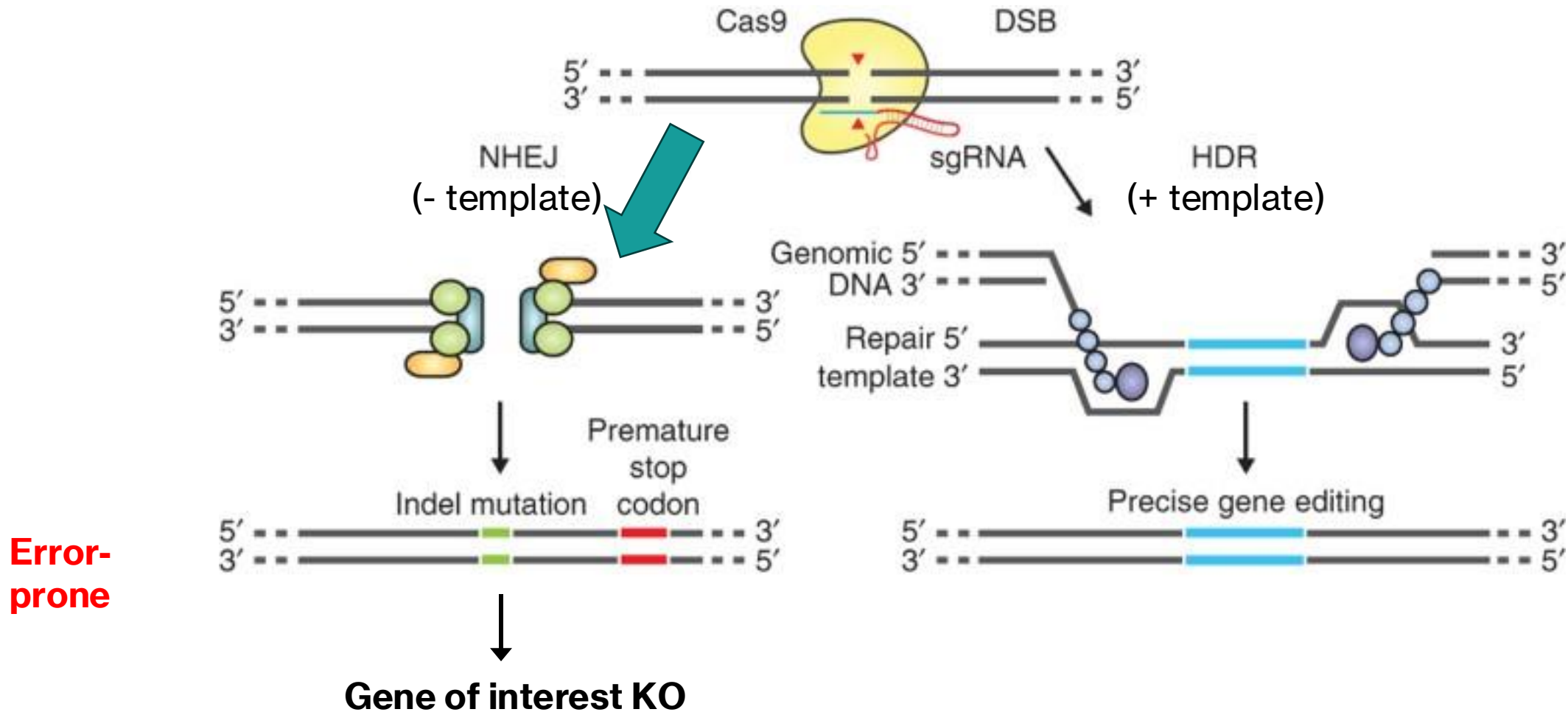
CRISPR guide design process and tools

Requirements for genome editing

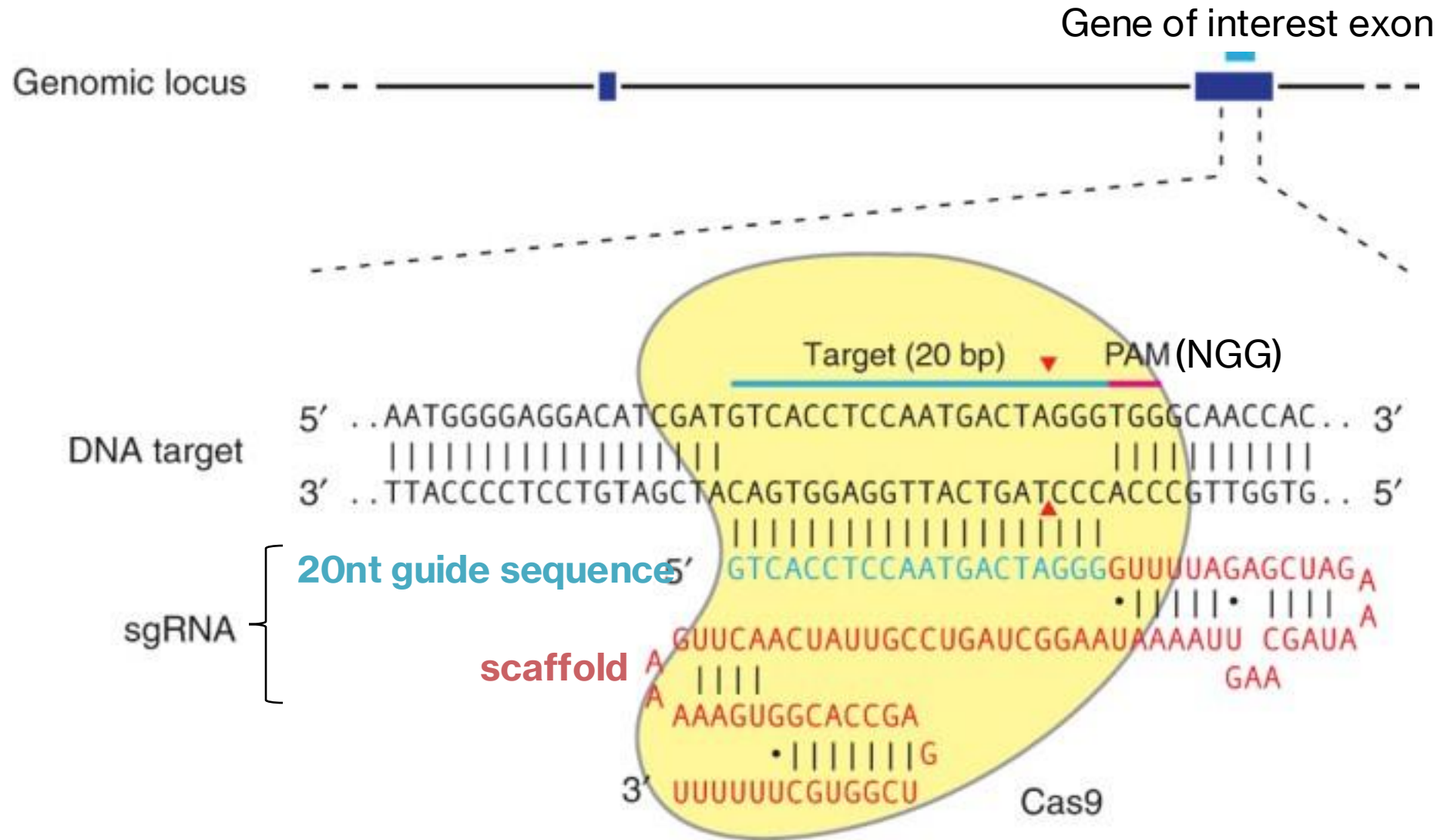
- Large Deletions
 - genes or loci
- Point Mutations
- Knockouts
 - NHEJ disruption
 - exon deletion
- Activation/Repression
 - CRISPRa/i



The CRISPR-Cas9 system

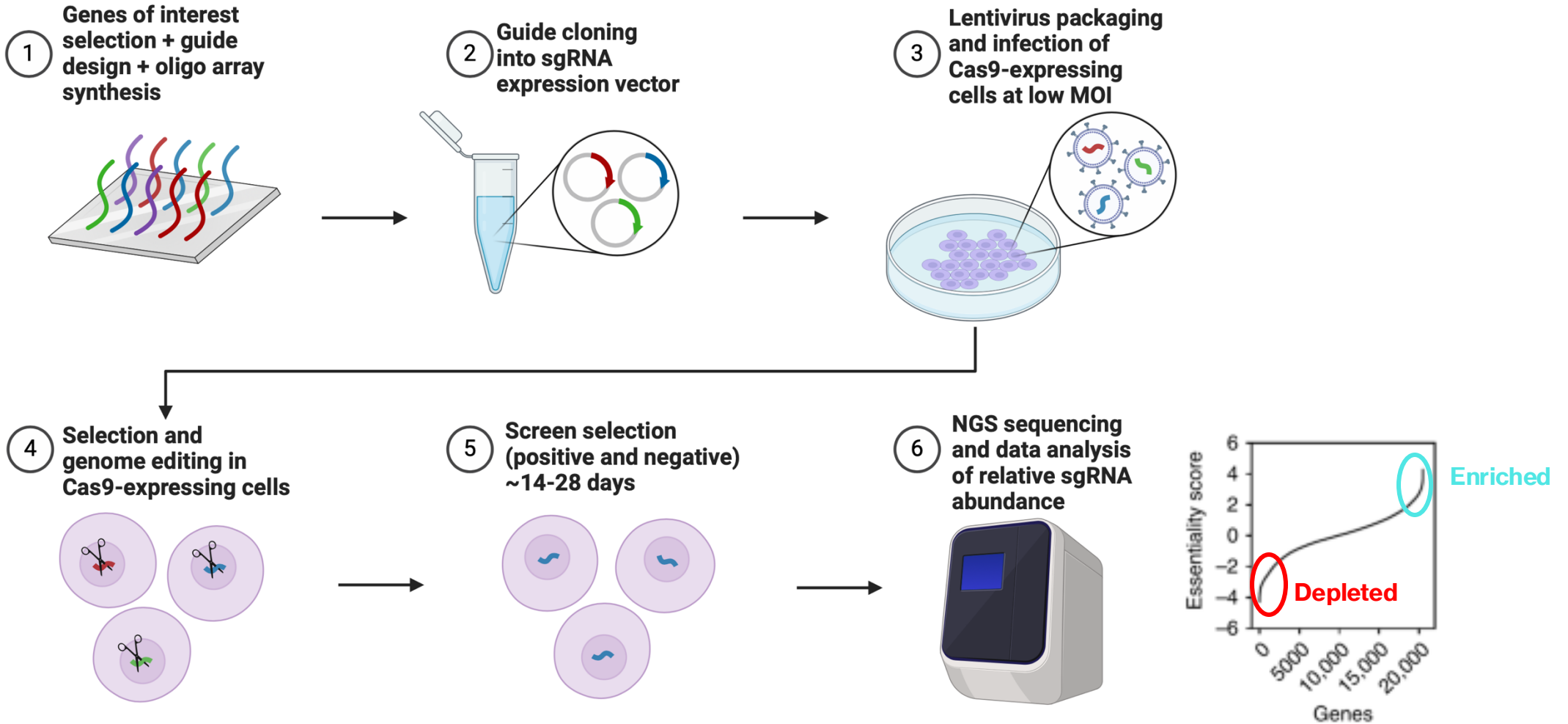


The CRISPR-Cas9 system

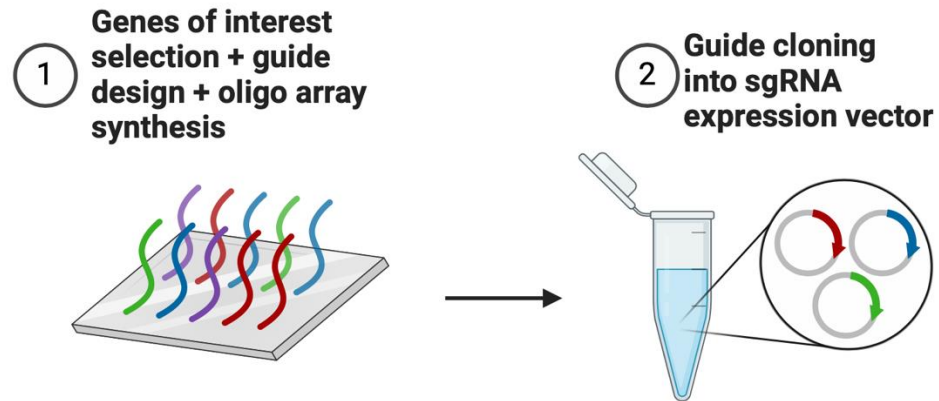


Adapted from Ran et al. Nature Protocols 2013

Pooled CRISPR-Cas9 KO screens with sgRNA libraries



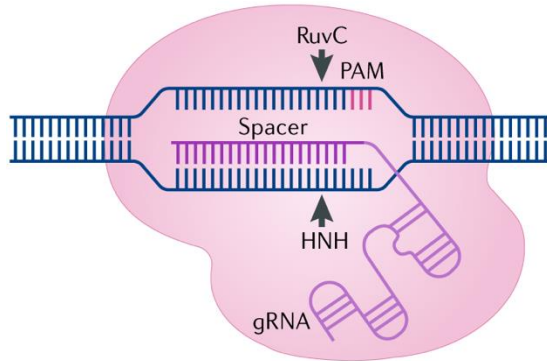
CRISPR-Cas9 sgRNA libraries



- At least 2 guides per gene to increase confidence in data and mitigate potential off-target effects (4 is recommended)
- Positive and negative control guides need to be included within each library
- Guide cloning: guide distribution should be as homogeneous as possible to improve data quality

Parallel approaches to CRISPR-Cas9 KO screening

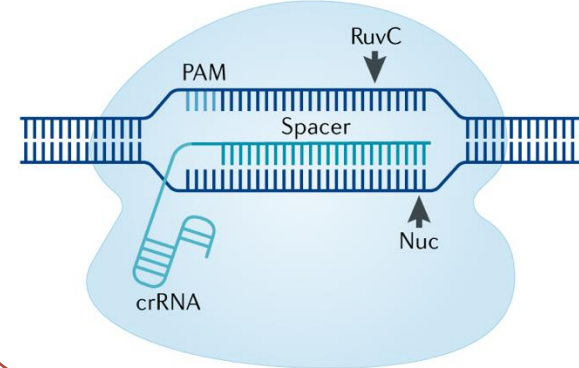
a Cas9 nuclease



Cas9

- Blunt DNA cut
- NGG PAM
- Guide length 20nt + scaffold
- Multiplexing possible with multiple promoters or tRNA
- Gene knock-out

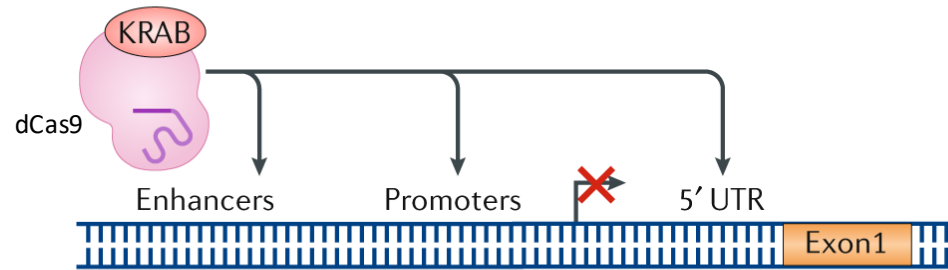
b Cas12a nuclease



Cas12a

- Staggered DNA cut with 5' overhang
- TTTV PAM
- Guide length 20-24nt + short constant section 20nt
- Multiplexing from a single RNA
- Gene knock-out

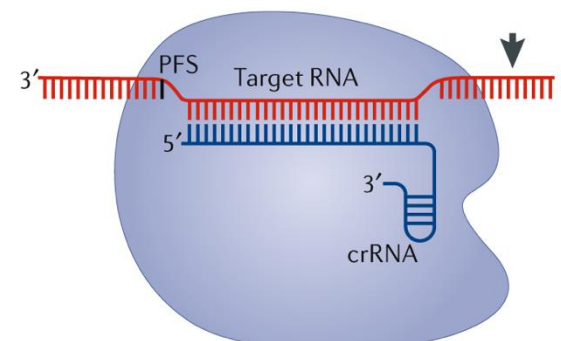
Transcription repression



CRISPRi

- Deactivated form of Cas9
- No DSB
- No permanent edits
- Reversible
- Represses gene expression
- Inhibition instead of knock-out

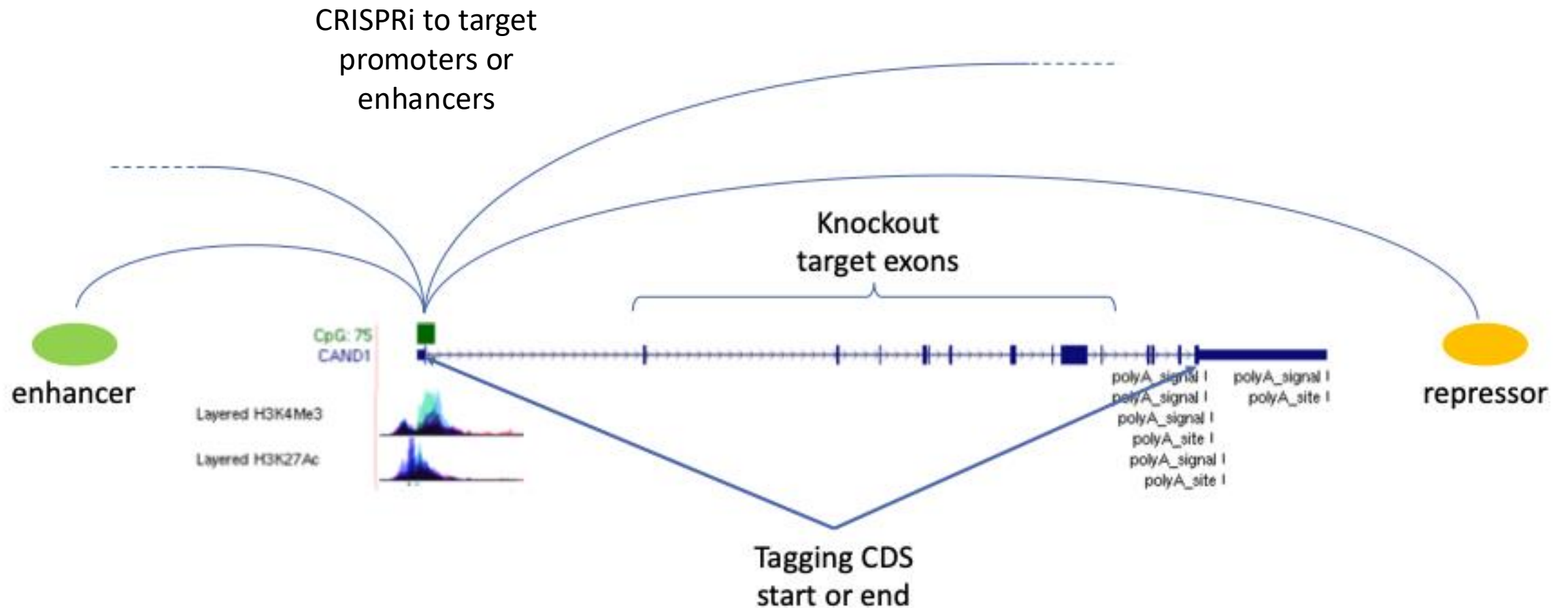
Cas13



Cas13d

- Smaller than Cas9
- No PAM required
- RNase activity of ssRNA
- 22nt guide + short conserved scaffold
- No permanent edits in genome
- Represses gene expression
- Knock-down instead of knock-out

Gene editing



Knockouts

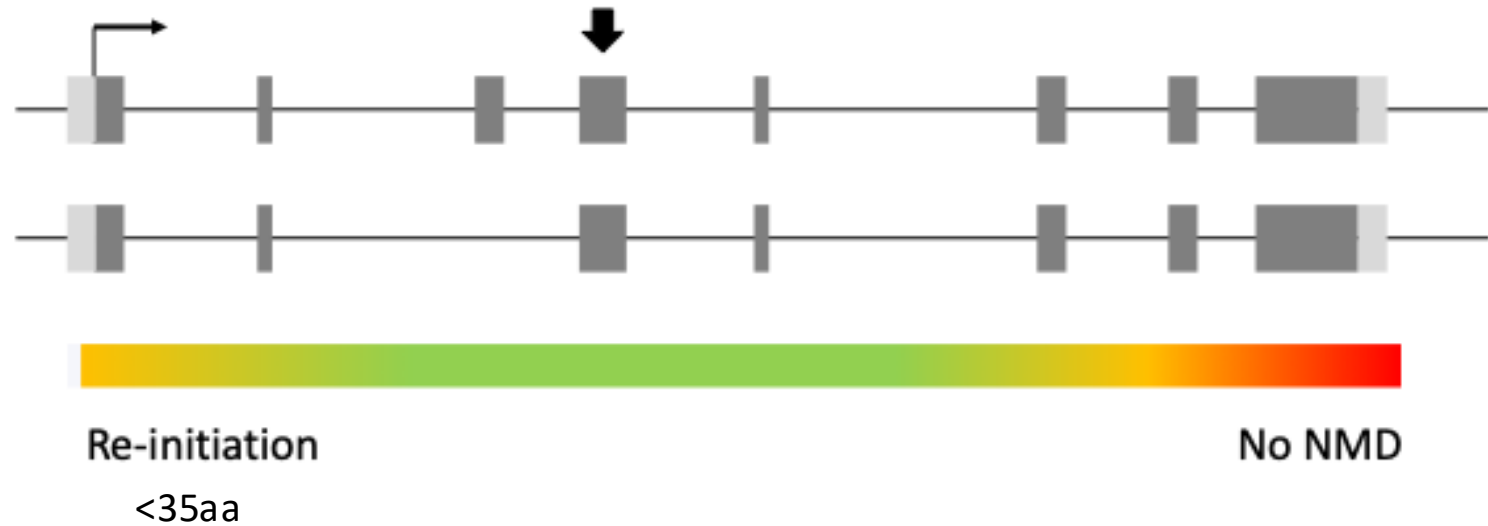
Typically knockout strategies involve:

- Insertion of targeted cassette
 - disrupts transcription and allows selection
- Frameshift mutations that induce NMD
 - Exonic indels – requires next-gen genotyping
 - Exon deletion – simple genotyping

Knockout – exon disruption

Targeted exons should ideally be:

- common to all isoforms
- within first 50% of protein
- avoid re-initiation
- induce NMD



guideRNA specificity

- Guide RNA typically consists of 20bp guide sequence adjacent to the nGG protospacer-adjacent motif (PAM).
- Base-pairing between guide sequences and complementary DNA sequences can occur despite multiple mismatches.
- 'Off-target' binding can reduce the target specificity and may result in off-target damage.
- Knowledge of potential off-target sites is therefore important.



Design Approaches

There are an increasing number of online CRISPR design tools, which employ different selection approaches.

Unbiased all gRNAs are included, which can be filtered

Biased gRNAs are pre-filtered to display 'optimal' designs.
Typically based on experimentally based algorithms
that score gRNA sequences.

Search methods are either:

Sequence-based requires an input sequence, typically up to 500nt

Gene-based uses gene name or Ensembl identifier

CRISPR design tools: Ensembl

Go back to the location tab and add a track to the “region in detail”.

Configure Region Image | Configure Overview Image | Configure Chromosome Image | Personal Data

Find a track

Active tracks
Favourite tracks
Track order
Search results

Sequence and assembly (3/29)
Sequence (2/4)
Simple features (1/3)
Clones & misc. regions (0/22)

Genes and transcripts (2/108)
Genes (2/4)
Long reads (0/17)
RNASeq models (0/87)

mRNA and protein alignments (0/2)
mRNA alignments (0/1)
Protein alignments (0/1)

Variation (2/10)
Sequence variants (1/3)
Failed variants (0/1)
Phenotype annotations (0/2)
Structural variants (1/4)

Regulation (1/166)
Regulatory features (1/1)
Activity by Cell/Tissue (0/161)
DNA methylation (0/2)
Other regulatory regions (0/2)

Comparative genomics (1/29)
Multiple alignments (0/4)
Conservation regions (1/4)
BLASTz/LASTz alignments (0/21)

Genome targeting (0/1)

Oligo probes (0/26)
Repeat regions (0/12)
Information and decorations (13/16)
Display options

Select from available configurations: Current unsaved

Genome targeting

☐ CRISPR SpCas9

Looking for more data? Search the [Trackhub Registry](#) for external sources of annotation

Key

Track style
Forward strand
Reverse strand
Favourite track
Track information

External data
User-added track

Please note that the content of external tracks is not the responsibility of the Ensembl project.
URL-based tracks may either slow down your Ensembl browsing experience OR may be unavailable.

CRISPR design tools: Ensembl

Guides targeting the forward strand are displayed on the top, and those targeting the reverse strand as shown below.



CRISPR design tools: Ensembl

Information and links to other sources are available for each guide.



CRISPR design tools: Ensembl

Individual CRISPR Summary

ID: 1166705895

[View in Genome Browser](#)

Species Human (GRCh38)
Location 19:44907839-44907861
Sequence CCAGCGCTGGGAACTGGCAC TGG
Strand +
Crispr in exon? Yes
Crispr in intron? No



Off-Target Counts	All	Exonic	Intronic	Intergenic
Total	227	31	103	93
Summary	{0:1, 1:0, 2:0, 3:21, 4:205}	{0:1, 1:0, 2:0, 3:6, 4:24}	{0:0, 1:0, 2:0, 3:8, 4:95}	{0:0, 1:0, 2:0, 3:7, 4:86}

Found 5 Related Crispr Pairs

[Show Crispr Pairs](#)

Off-Target Sites

Note: the row highlighted in blue is the original CRISPR

WGE ID	Location	Sequence	Mismatches ⓘ	Strand	Type
1166705895	Original CRISPR	CCAGCGCTGGGAACTGGCAC TGG			Exonic
1166705895	19:44907839-44907861	CCAGCGCTGGGAACTGGCAC TGG	0	+	Exonic
1183733406	22:39630637-39630659	GCAGAGCCGGGAACTGGCAC AGG	3	+	Intronic
1021787321	7:24164871-24164893	GCAGCTCTAGGAACTGGCAC TGG	3	+	Intergenic
1001791760	5:174463776-174463798	CCACCCTGGCAACTGGCAC AGG	3	+	Intergenic
1119608739	14:76043865-76043887	CCAGGGCTTGCAACTGGCAC TGG	3	-	Intronic



CRISPR design tools: WGE

<https://wge.stemcell.sanger.ac.uk>

HTGT WGE

Home

CRISPR Finder ▾

Help ▾

About

Contact

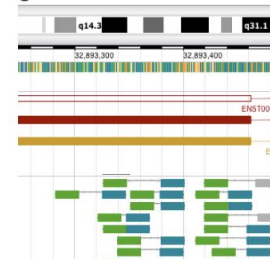


Wellcome Sanger Institute Genome Editing (WGE) is a website that provides tools to aid with genome editing of human and mouse genomes

[CRISPR Finder](#) [Ensembl for Human](#) [Ensembl for Mouse](#)

The CRISPR Finder will show CRISPR sites (paired or single) in and around genes. You can ask the finder to score the pairs for potential off-target sites, and browse individual and paired CRISPR sites using the Genoverse genome browser tool. We also provide the ability to find CRISPRs in genomic sequence or by gRNA:

Find CRISPRs in our genome browser:



Find CRISPRs by gene using our table:

Pair			
Exon ID	Spacer	Status	Summary
ENSE00003666217	20	Complete	closest: None total_pairs: 1 max_distance: 1000
	3	Complete	closest: None total_pairs: 1 max_distance: 1000

Find CRISPRs by 20bp gRNA:

Sequence

Species ☐ Human (GRCh37) ☒ Human (GRCh38) ☐ Mouse (GRCm38)

Ensembl In gene In e:

Find CRISPRs in genomic sequence:

IAAGGAATGTTCCCAATAGTAGACATAAAAGTCTTCG

Crispr ID	Ensembl	In g
1106710403	13:32325087-32325109	No
1106710404	13:32325088-32325110	No
1106710405	13:32325110-32325132	No

Find off-targets by sequence:

☐ Mouse (GRCm38)

Orientation ☐ PAM Right (NGG) ☒ PAM Left (CCN)

Sequence	Ori
GTGTCACTGAACTTACTCT	par
GTGTCCCAGAACTTACTCT	par

If you use this site in your research, please cite:

WGE: A CRISPR database for genome engineering. Alex Hodgkins; Anna Farne; Sajith Perera; Tiago Grego; David J. Parry-Smith; William C. Skarnes; Vivek Iyer (Bioinformatics 2015)
doi:10.1093/bioinformatics/btv308

CRISPR design tools: Vienna Bioactivity Score (VBC)

<http://vbc-score.org>



Species
Homo sapiens (hg38) ▼
Choose a species

Gene name ▼
Select one or multiple gene names

☐ Enter sequence

SEARCH

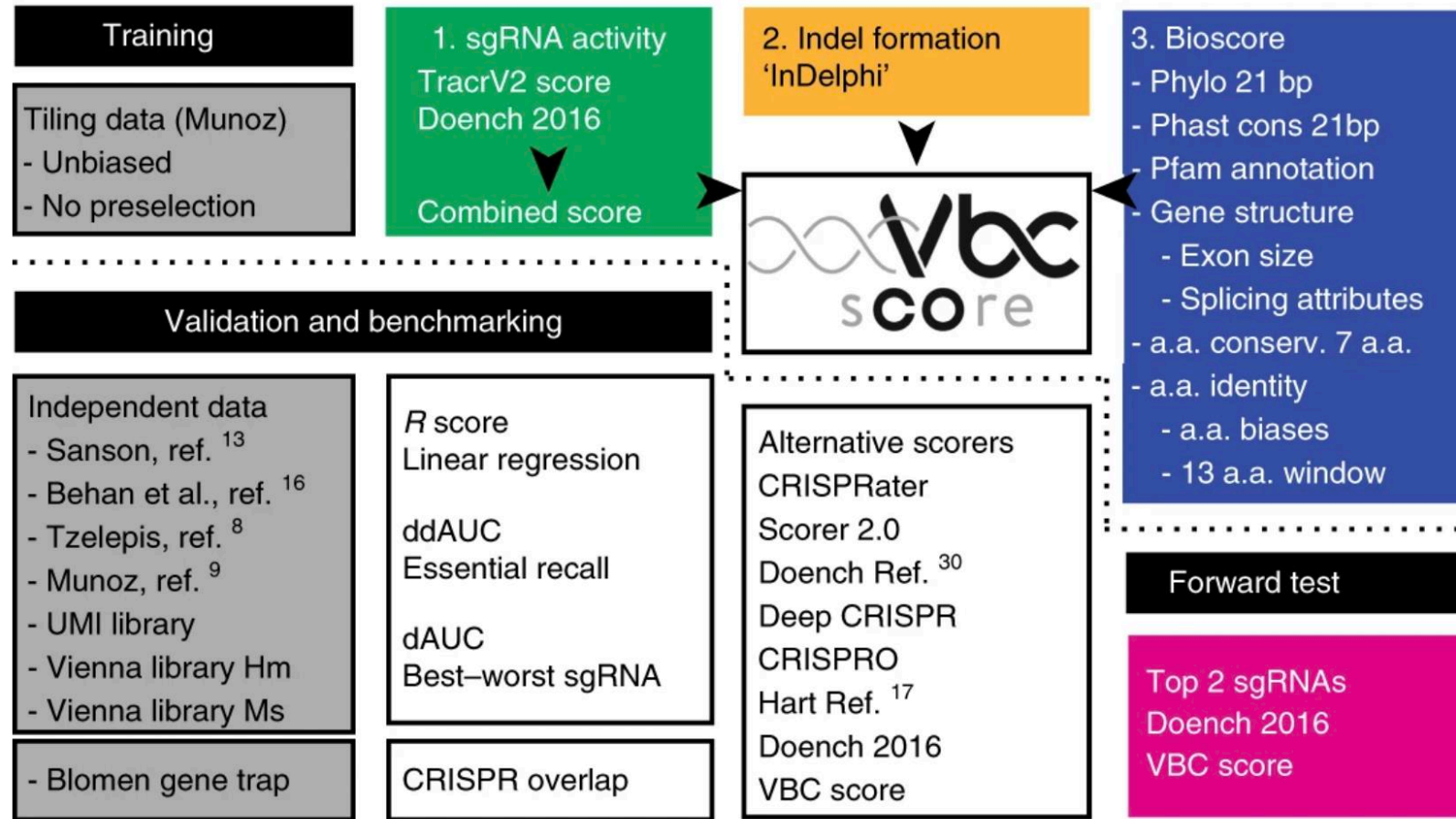
Article | Published: 08 June 2020

Multilayered VBC score predicts sgRNAs that efficiently generate loss-of-function alleles

[Georg Michlits](#), [Julian Jude](#), [Matthias Hinterndorfer](#), [Melanie de Almeida](#), [Gintautas Vainorius](#), [Maria Hubmann](#), [Tobias Neumann](#), [Alexander Schleiffer](#), [Thomas Rainer Burkard](#), [Michaela Fellner](#), [Max Gijssbertsen](#), [Anna Traunbauer](#), [Johannes Zuber](#) ✉ & [Ulrich Elling](#) ✉

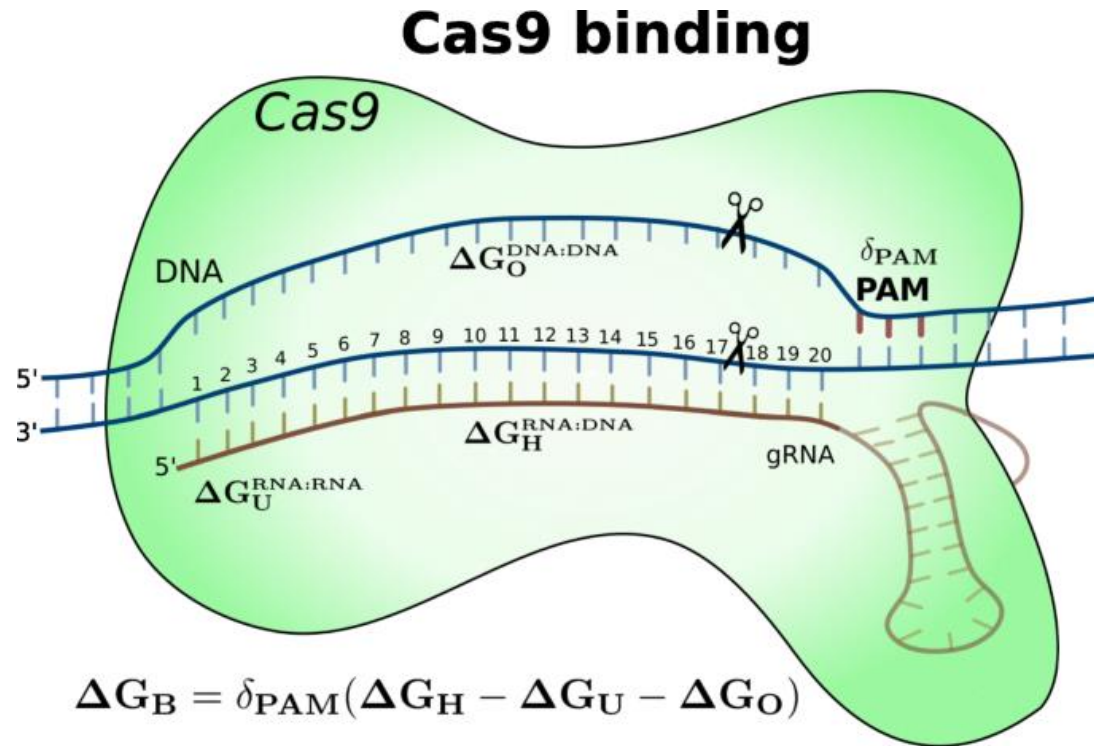
[Nature Methods](#) **17**, 708–716 (2020) | [Cite this article](#)

CRISPR design tools: Vienna Bioactivity Score (VBC)



Michlits et al. Nat Methods 2020

CRISPR design tools: CRISPR off/spec



Uses binding energies to determine guide off targets and binding specificities

<https://rth.dk/resources/crispr/crisproff/>

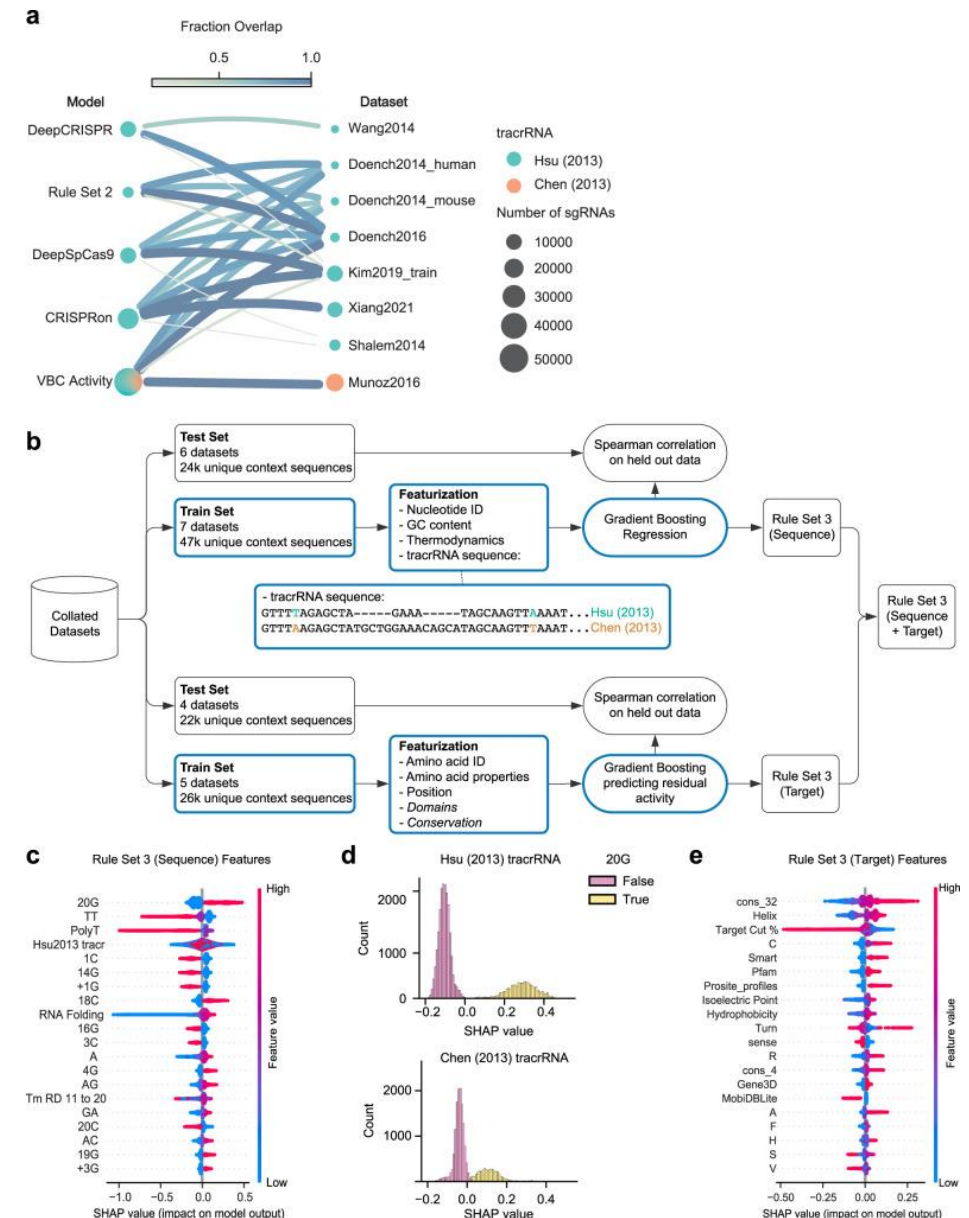
CRISPR design tools: on target efficacy score- rule set 3

Article | [Open access](#) | Published: 06 September 2022

Accounting for small variations in the tracrRNA sequence improves sgRNA activity predictions for CRISPR screening

[Peter C. DeWeirdt](#), [Abby V. McGee](#), [Fengyi Zheng](#), [Ifunanya Nwolah](#), [Mudra Hegde](#) & [John G. Doench](#)

[Nature Communications](#) **13**, Article number: 5255 (2022) | [Cite this article](#)



CRISPR design tools: CRISPick

<http://portals.broadinstitute.org/gppx/crispick/public>

CRISPick Ranks and picks candidate CRISPRko/a/i sgRNA sequences to maximize on-target activity for target(s) provided.



Last update: 2025-06-25 | [Changelog](#)

NEW (2024-09-24)! For the GRCh38 genome, CRISPick has added support for [gnomAD](#) variation filtering. This feature is enabled by default, and deprioritizes selection of sgRNAs that overlap sites of variation, e.g. SNPs and indels. See the "Advanced Options" section below for details.

Reference Genome

- ☒ Human GRCh38 (NCBI RefSeq v.GCF_000001405.40-RS_2024_08)
- ☐ Human GRCh38 (Ensembl v.113)
- ☐ Human GRCh37 (NCBI RefSeq v.GCF_000001405.25-RS_2024_09)
- ☐ Mouse GRCm38 (NCBI RefSeq v.108.20200622)
- ☐ Mouse GRCm38 (Ensembl v.102)
- ☐ Rat mRatBN7.2 (NCBI RefSeq v.GCF_015227675.2-RS_2023_06)
- ☐ Rat mRatBN7.2 (Ensembl v.113)

Mechanism

- ☒ CRISPRko
- ☐ CRISPRa
- ☐ CRISPRi

Enzyme

- ☒ SpyoCas9 (NGG)
- ☐ [Chen \(2013\)](#) tracrRNA
- ☐ [Hsu \(2013\)](#) tracrRNA
- ☐ SaurCas9 (NNGRR)
- ☐ AsCas12a (TTTV)
- ☐ enAsCas12a

Full-genome CRISPick downloads

How CRISPick works

FAQ

Target(s)

- ☒ Quick gene lookup
- ☐ Bulk/Advanced targets
- ☐ Upload file

Copy IDs

Accepted target formats

ID

Gene Symbol

CDC5L, Brca1

Sequence

Raw

TTGTAGCATCGCAGGTAGCAAACAGTTACTAGG

Coordinates

Point

NC_000001.11:+:127140001

Task1: Design guides to knock-out the human MYC gene using CRISPR-Cas9

1. Design 5 sgRNAs against *MYC* using Ensembl genome browser
2. Use nucleotide blast on the Ensembl webpage to blast the selected guide sequences- do they have any off targets?
3. Search VBC database for your selected guides: what are their VBC scores? How do these scores compare to the best scoring guides according to VBC?
4. Use CRISPick web tool to pick 5 guides for the same gene- do any of these overlap with the ones you selected?
5. Use nucleotide blast on the Ensembl webpage to blast the best guide sequence selected by CRISPick (Pick order 1 on the output file)- does it have any off targets?
6. Search the location of this guide on the Ensembl genome browser- which exon does it target?



Thank you for your
attention!